👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 1 · STANDARD 5.MD.C.5

Math is Explaining and Sharing

Lesson 1-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can construct an argument using equations, drawings, or words, and use volume to defend a recommendation.

Language Objective: I can explain my thinking with clear math vocabulary using the words argument, conjecture, and cubic units.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

argument · conjecture · counterexample · volume · cubic unit

☝️ *Tap any word to see its meaning and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. A chain of reasons used to defend your thinking is an _____.

2. A statement you believe is true but have not yet proven is a _____.

3. A single case that shows a statement is not always true is a _____.

4. The amount of space a solid figure takes up, in cubic units, is its _____.

5. A cube with edges 1 unit long, used to measure volume, is a _____ _____.

Write About the Math

How can the group determine the amount of soil needed, and how can they check that their answer is accurate?
¿Cómo puede el grupo determinar la cantidad de tierra necesaria, y cómo pueden comprobar que su respuesta es exacta?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **argument** *argumento*
- ☐ **conjecture** *conjetura*
- ☐ **counterexample** *contraejemplo*
- ☐ **volume** *volumen*
- ☐ **cubic unit** *unidad cúbica*

Model: *Three large boxes hold 240 × 3 = 720 cubic inches, and two jumbo boxes hold 400 × 2 = 800 cubic inches, so 800 > 720.* — A strong answer calculates both totals (720 and 800) and compares them to recommend the jumbo boxes. A weak answer picks an option without computing and comparing both volumes.

Start

Complete the frames. Use the word bank.
Completa las oraciones. Usa el banco de palabras.

- **To find how much soil is needed, find the ____ of the garden beds. Because volume covers three dimensions, label the answer in ____ feet. Finding the volume two different ways and getting matching totals shows the solution is ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.
Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.

- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** argument
2. **Notes 2:** conjecture
3. **Notes 3:** counterexample
4. **Notes 4:** volume
5. **Notes 5:** cubic unit
6. **Watch for this mistake:** *Reporting volume without cubic units — saying the garden needs '48 feet' of soil, which is a length, or '48 square feet', which is an area.*
7. **Try It 1:**
8. **Try It 2:**